

Topic 5: Arms Race (Group 10)

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Overview



Scenario: Purple (p) and Green (g) are two countries competing in an arms race.

Key Factors:

- More inclined to increase defence spending as the other's rises
- Less inclined to increase spending as its own spending rises
- How does public sentiment affect defence spending?
- How do limited resources affect defence spending?

Research Question: What happens to their arms spending over time?

Motivation: It helps predict military escalation between countries, which can inform policies to promote peace and prevent conflict.



Base Model

Model Equations

The equations for the base model are:

$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r \quad (1)$$

$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s \quad (2)$$

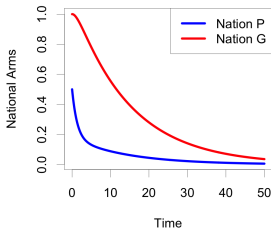
- Feasibility Condition: $(p(t), g(t) \geq 0)$
- a, c : Mutual Fear ($a, c > 0$)
- b, d : Spending Burden ($b, d > 0$)
- $(r, s) < 0$ for trust and $(r, s) > 0$ for hostility

Long Term Behavior ($r = s = 0$)

Assuming no underlying trust/hostility, we have:

$$bd > ac$$

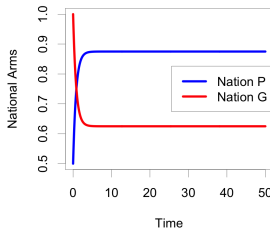
$$a=0.1, b=0.7, c=0.2, d=0.1$$



Complete mutual disarmament

$$bd = ac$$

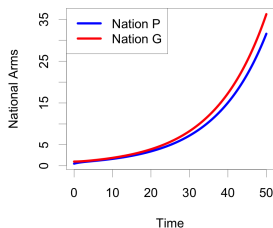
$$a=0.7, b=0.5, c=0.5, d=0.7$$



Stabilized spending

$$bd < ac$$

$$a=0.5, b=0.5, c=0.2, d=0.1$$



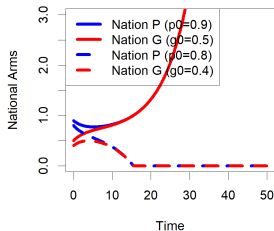
Perpetual arms race



Long Term Behavior with Trust/Hostility

$$bd < ac, r < 0, s < 0$$

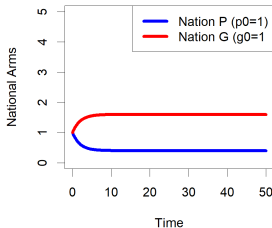
$$a=c=0.25, b=d=0.1, r=s=-0.1$$



Arms equilibrium
determined by initial
spending

$$bd = ac, r < 0, s > 0$$

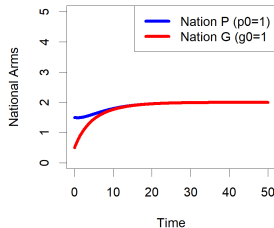
$$a=b=c=d=0.25, r=-0.3, s=0.3$$



Stable arms spending

$$bd > ac, r > 0, s > 0$$

$$a=c=0.1, b=d=0.25, r=s=0.3$$



Stable arms spending



Mutual Logistic Fear

Limitation of the Current Model: Each country has a limited amount of resources at a given time.

Solution: Introduce self-limiting logistic growth.

Model Equations

The equations after adding logistic growth are:

$$\frac{\partial p}{\partial t} = a \left(1 - \frac{p(t)}{K_p} \right) g(t) - bp(t) + r \quad (3)$$

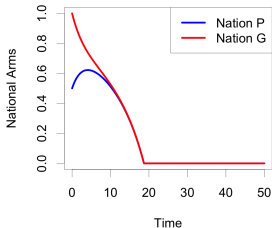
$$\frac{\partial g}{\partial t} = c \left(1 - \frac{g(t)}{K_g} \right) p(t) - dg(t) + s \quad (4)$$



Long-Term Behavior of Mutual Logistic Fear

$$bd < ac, r < 0, s < 0$$

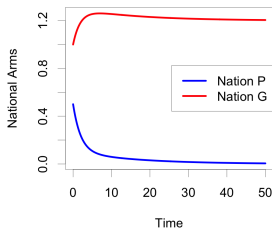
$a=0.25, b=0.1, c=0.25, d=0.1, r=-0.1, s=-0.1, Kp=5, Kg=5$



Mutual disarmament

$$bd = ac, r < 0, s > 0$$

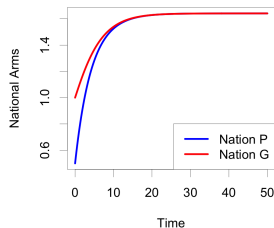
$a=b=c=d=0.25, r=-0.3, s=0.3, Kp=5, Kg=5$



Stabilized spending

$$bd > ac, r > 0, s > 0$$

$a=0.1, b=0.25, c=0.1, d=0.25, r=0.3, s=0.3, Kp=5, Kg=5$



Stabilized spending



Conclusion and Next Steps

Conclusion

- **Base Model:** In this model, long-term outcomes depend only on combined fear versus combined spending preservation.
- **Public Sentiment:** Initial spending influences this model's behavior and shifts the equilibrium which makes stable positive spending more likely.
- **Mutual Logistic Fear:** This model accounts for economic limitations and prevents unending arms race with infinite spending.

Next Steps

- Sensitivity Analysis
- Analytical solutions for equilibrium and stability

Thank you for listening!



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